

Using Pre-visualization to bring HBO's "The Last Of Us" To Life

Micah Malinics

SOKRISPYMEDIA, Philadelphia, PA
micah@sokrispymedia.com

Sam Wickert

SOKRISPYMEDIA, Greenville, SC
sam@sokrispymedia.com

Brendan Forde

SOKRISPYMEDIA, Greenville, SC
brendan@sokrispymedia.com

ABSTRACT

During 2021, our team (SOKRISPYMEDIA) was brought on as a vendor to begin work on HBO's "The Last Of Us" TV adaptation. During our work on the show, we utilized the latest virtual production technology to produce a previsualization process that allowed us to capture and playback the 3D previsualization in real time over Zoom, allowing the creative stakeholders to provide real time input on camera movement, character movement, camera lens settings and much more.

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1 OVERVIEW

In early 2021 we produced this YouTube educational video showcasing the cutting edge developments of real time rendering technology, real time camera tracking, and motion capture, and how the collision of all three of those will allow for opportunities to produce 3D visual content in real time that was never possible before.

We utilized consumer-grade motion trackers from HTC, a Blackmagic Design Decklink 8K to I/O a camera feed inside of Unreal Engine, and a consumer-grade motion capture suit to input real-time motion data inside of the software, allowing us to pilot a virtual camera inside of Unreal Engine and control one or more virtual characters using motion capture. With this, we could record either scenes entirely within Unreal Engine, or real-life composites of virtual characters against real world footage.

After demonstrating these capabilities, we were brought on to work on HBO's latest TV show to produce a previsualization of a complex sequence for the pilot episode, involving very precise camera movement and car stunts.

We built an environment inside of Unreal Engine to match the real world city the show was filming in. From there, we utilized this same real-time virtual production technology to record real-life camera movement and character movements inside of the 3D previsualization scene for the show.

Additionally, and most importantly, once we had the initial blocking in place, we were able to feed the same virtual camera input into video conferencing software and present the show's creative team a preview of the storyboarded sequence running in real time. Since

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Figure 1: Behind The Scenes of the real time capture.



Figure 2: Direct camera output of the real time motion capture, camera track, and composite.

the output of the virtual camera was running in real time, we were able to make "on the fly" changes to the camera and scene, allowing us to move the camera closer or further, adjust lens and exposure settings, and even re-block the scene entirely. Additionally, since we had one of our team members setup to be operating motion capture inside the scene, we were even able to make adjustments to the actor's performance in real time.

With the combination of all of this new virtual production technology, we were able to define a pipeline that the show's team had never seen before. When we are able to problem-solve and audition new camera and character movements in real time, we were able to produce and bring the sequence to final on a very truncated timeline and without the need to setup any in-person feedback or review sessions (which was particularly important during the pandemic).

Lastly, since we were able to dial in the sequence to the exact precision to the creative team's requests, the resulting sequence we designed in previsualization nearly matched identically to the final sequence filmed for the show.